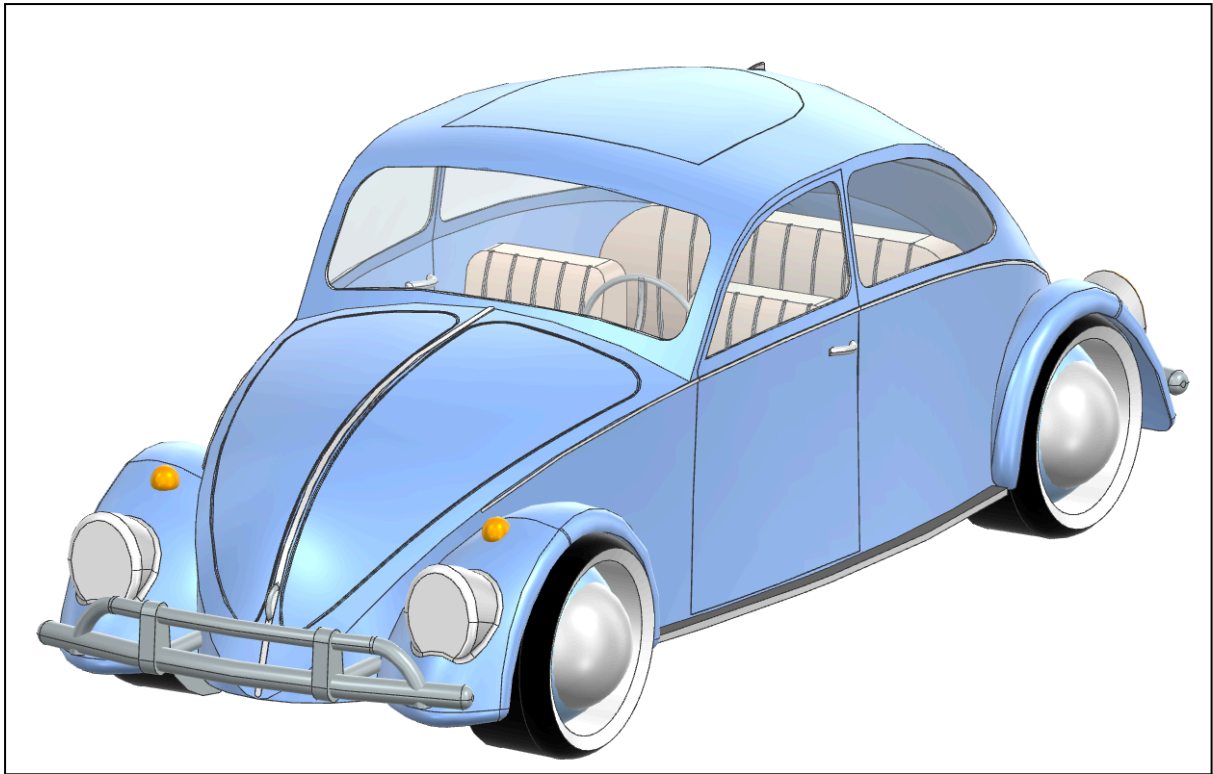


ME4042 - Interactive CAD and CAE

Lab-Based Midterm – Classic VW Beetle



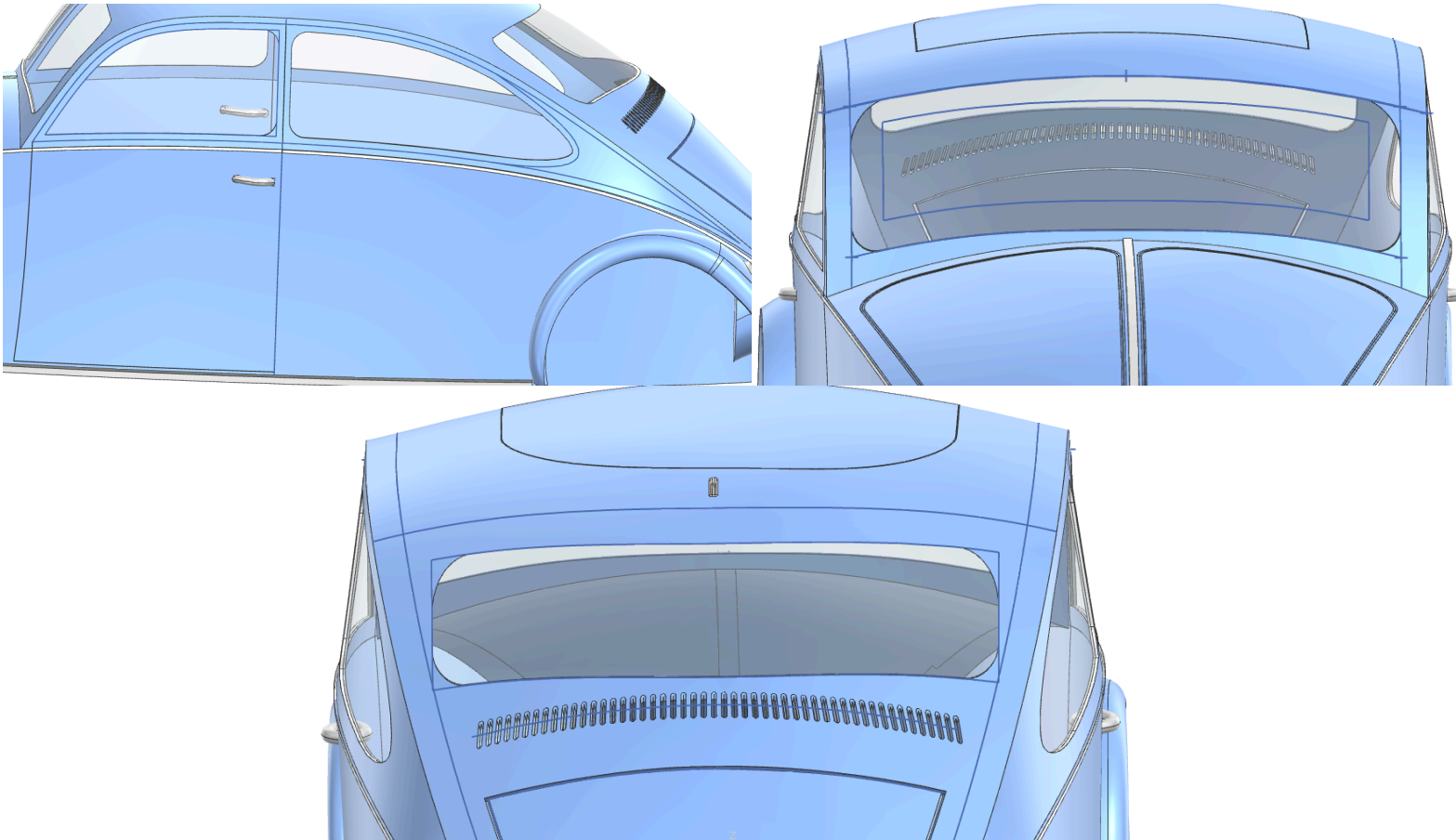
João Pedro Dudziak Fonseca

Submitted to: Dr. Suresh K Sitaraman

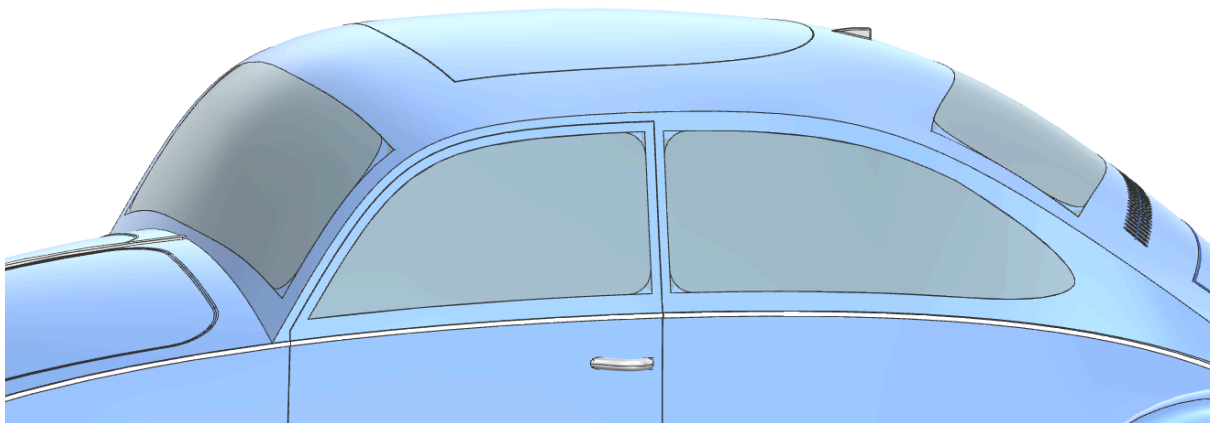
TA: Jason Daniel Schwartz, Liliana Claire Whitesell I

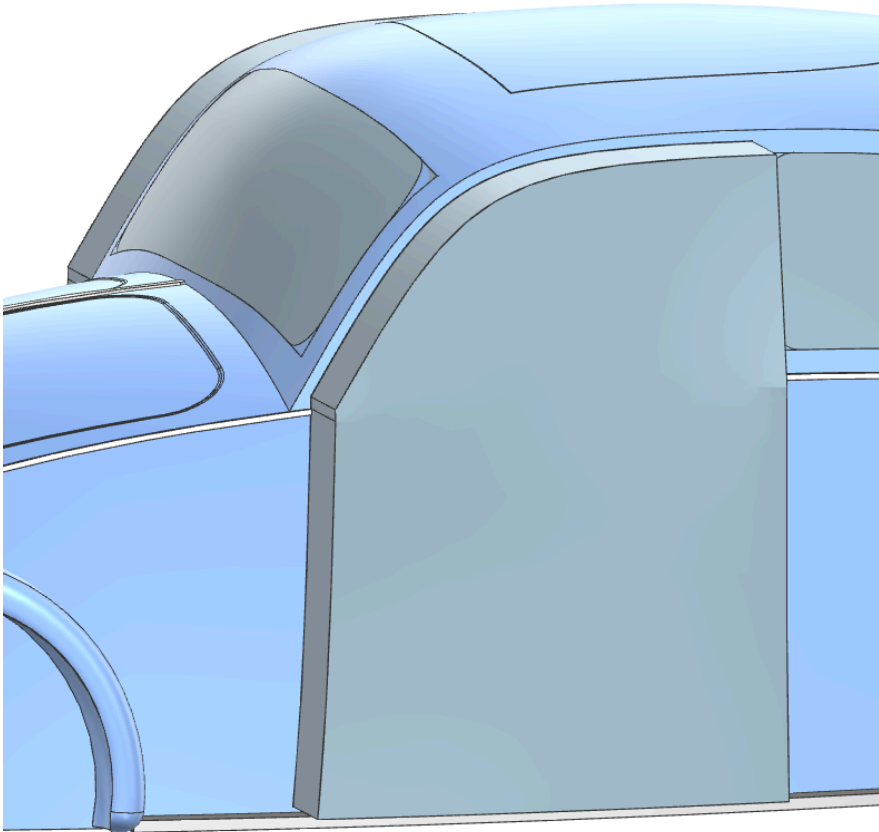
Date: 10/21/2025

Modelling Process



After creating the main geometry through the instructions, I drew a series of “offset in face” lines, splines, as well as straight lines defined with endpoints from other curves projected on the face in order to make the outline of the windows and doors.

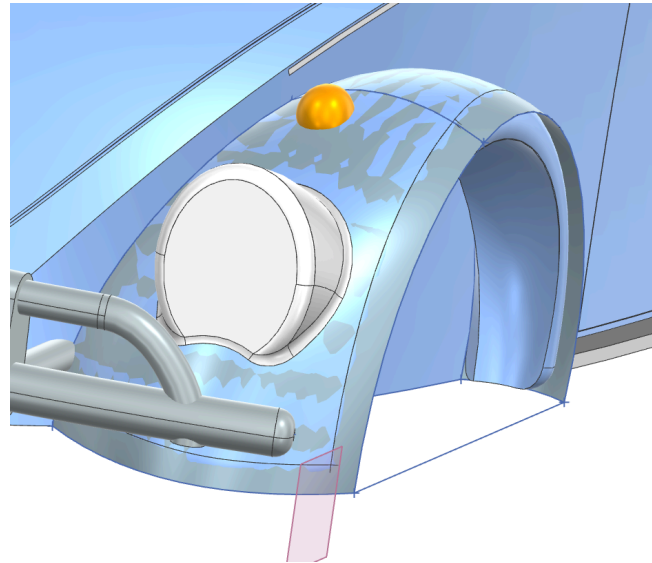
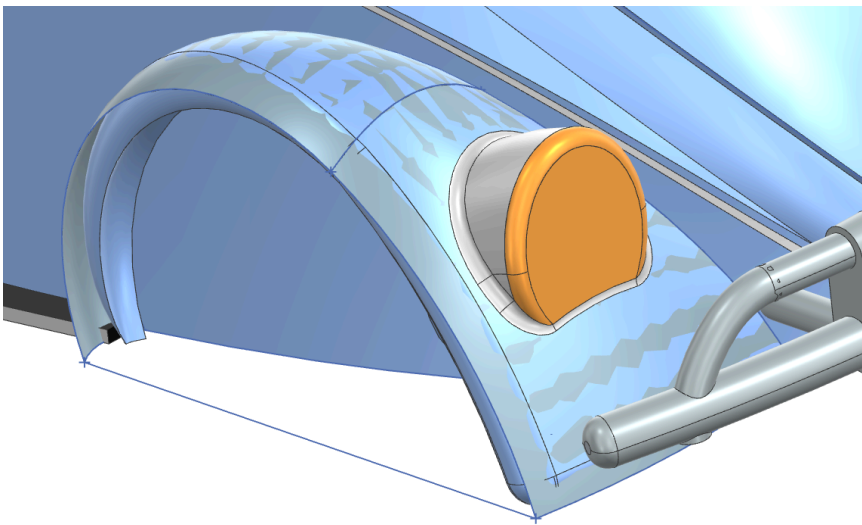




Then, I extruded all of these curves through the car, used "Fill Surface" on their ends and combined them to make a solid. I then subtracted each of these solids from the main body, making the holes for the windows and doors, making the doors as separate bodies but from the same geometry, but thickened by 1mm inward.

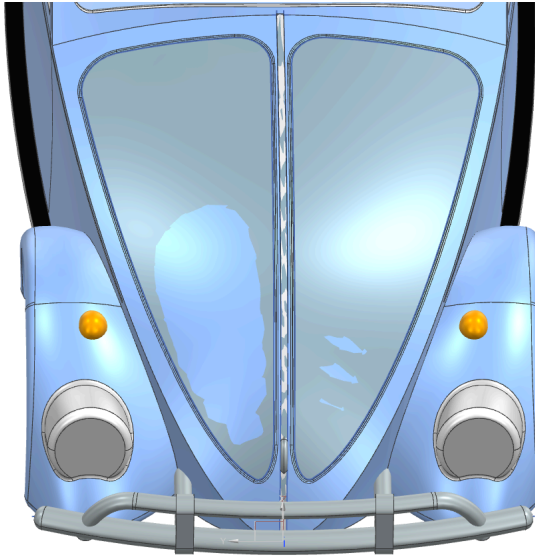
I then added edge blends on the edges to make the design look closer to the windows of the beetle.

After that, I retraced the lines of the windows at their width middle to create the transparent windows, later combining them to the husk/doors.



After that, I began creating the front and back wheel covers. I created a datum plane and drew the sketch of the covers, projected that drawing to the husk as well and created three connecting splines.

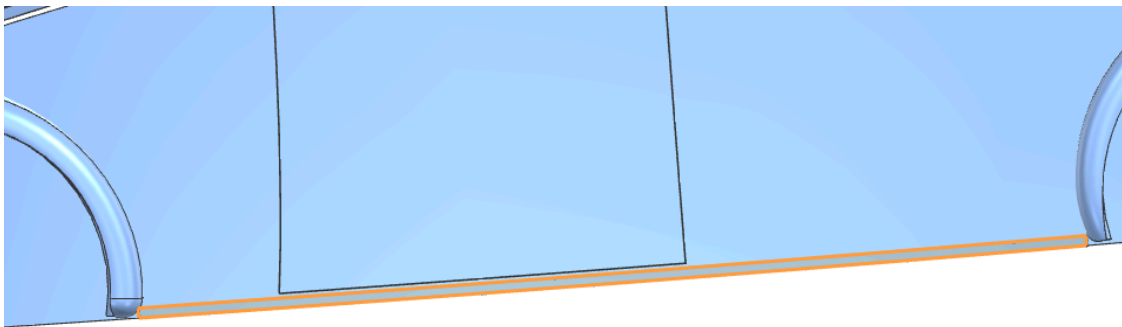
With that, I created surfaces of the covers, thickened them and added edge/face blends to make them more similar to the original. I then mirrored the front and back covers (which are different) to the other side of the husk and combined everything.



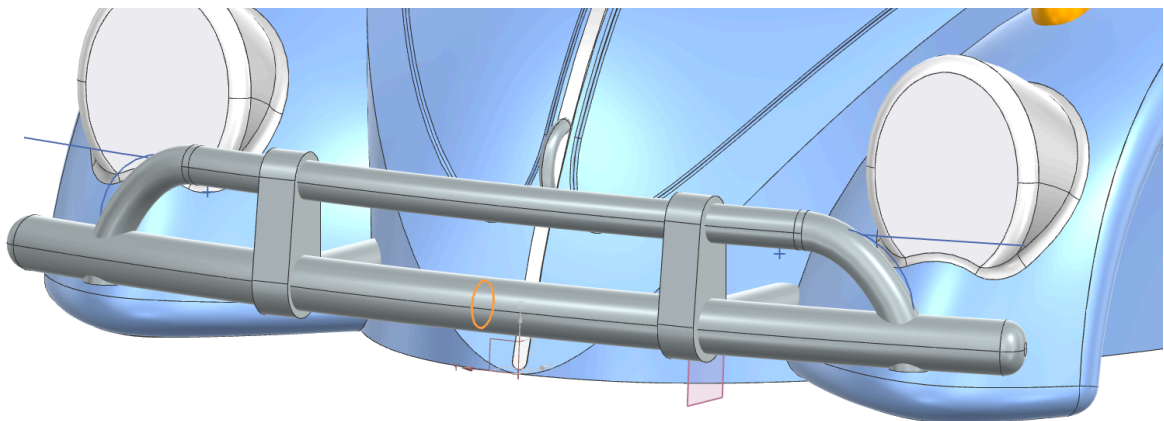
I then drew an outline of the hood features (side “carves” and the white line in the middle) on the bottom plane - using Offset Curves, blends and splines. I then projected that sketch onto the hood.

With that, for the dents, I created solids by Extruding and Fill Surface combining and subtracting them from the hood. For the line, I extruded the sketch upward and did the same process, but combined instead of subtracted.

I then added chamfers/face blends to the edge of the dents to make the gradual transition seen in the original car.



I then moved on to the side steps. Creating a drawing in y-plane, I projected that onto the husk, extruded it, fill surfaced, created the solid, mirrored it and combined it to the husk.



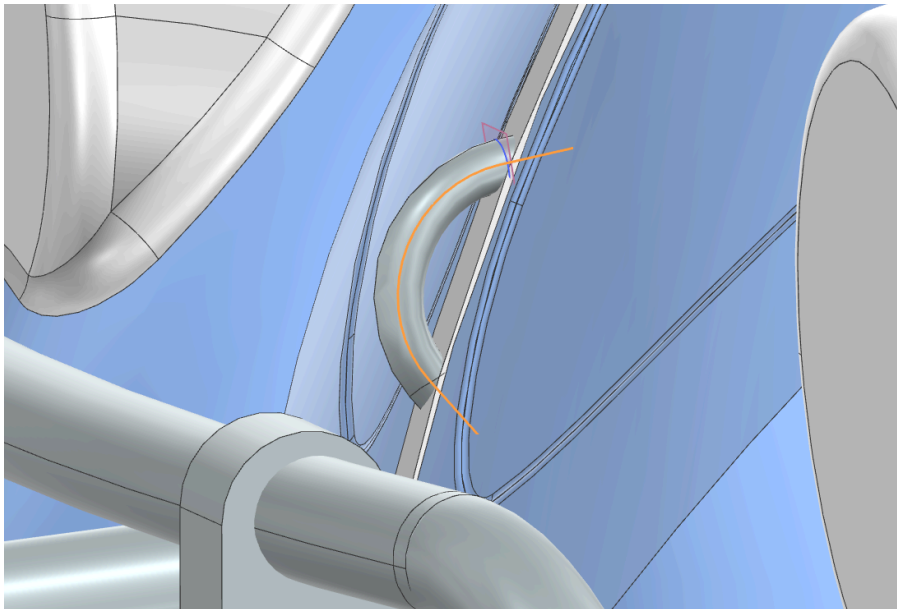
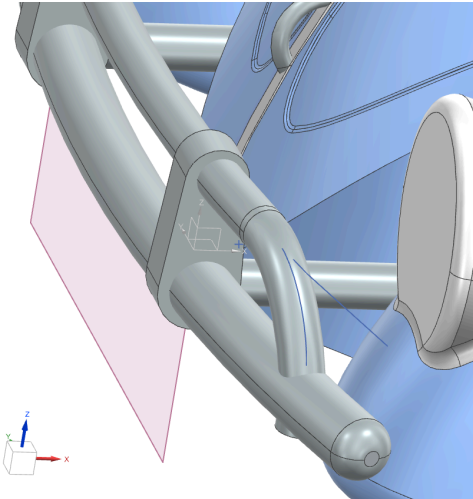
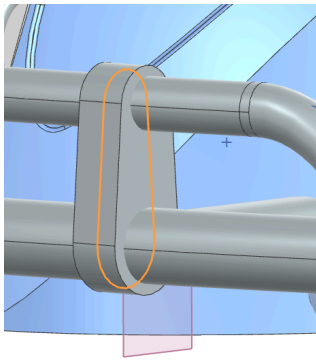
Then for the front bumpers, I made a spline from points I created in front of the car with a curved silhouette and a circle at the middle of it. With them, I created a surface and then thickened it. I then edge blended the ends to make them look like bumpers.

After, I offset the same spline upwards to make the top part of the bumpers. I again drew the guide circle but with a smaller radius, made the surface and thickened it. I only extruded it up to a point on the lower bumper through a point I arbitrarily drew as well. After, I drew a guide

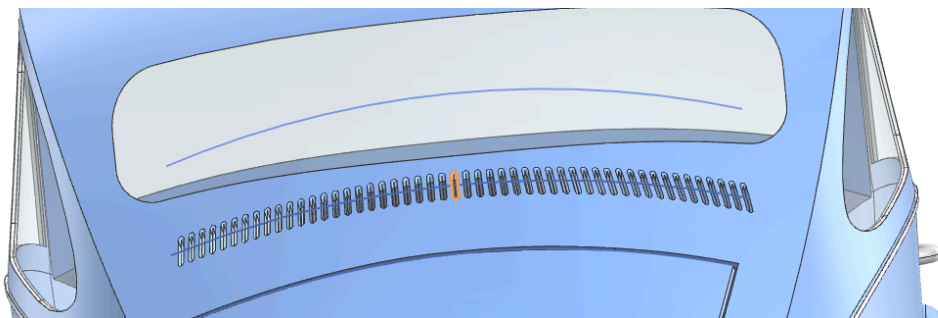
semi circle connecting the top and bottom splines and created the connection with the same method, later mirroring it to the other side.

Then, for the middle fixtures, I created a datum and drew the shape on it, extruding as a solid and combining it with the top and bottom bumpers (later mirroring it as well).

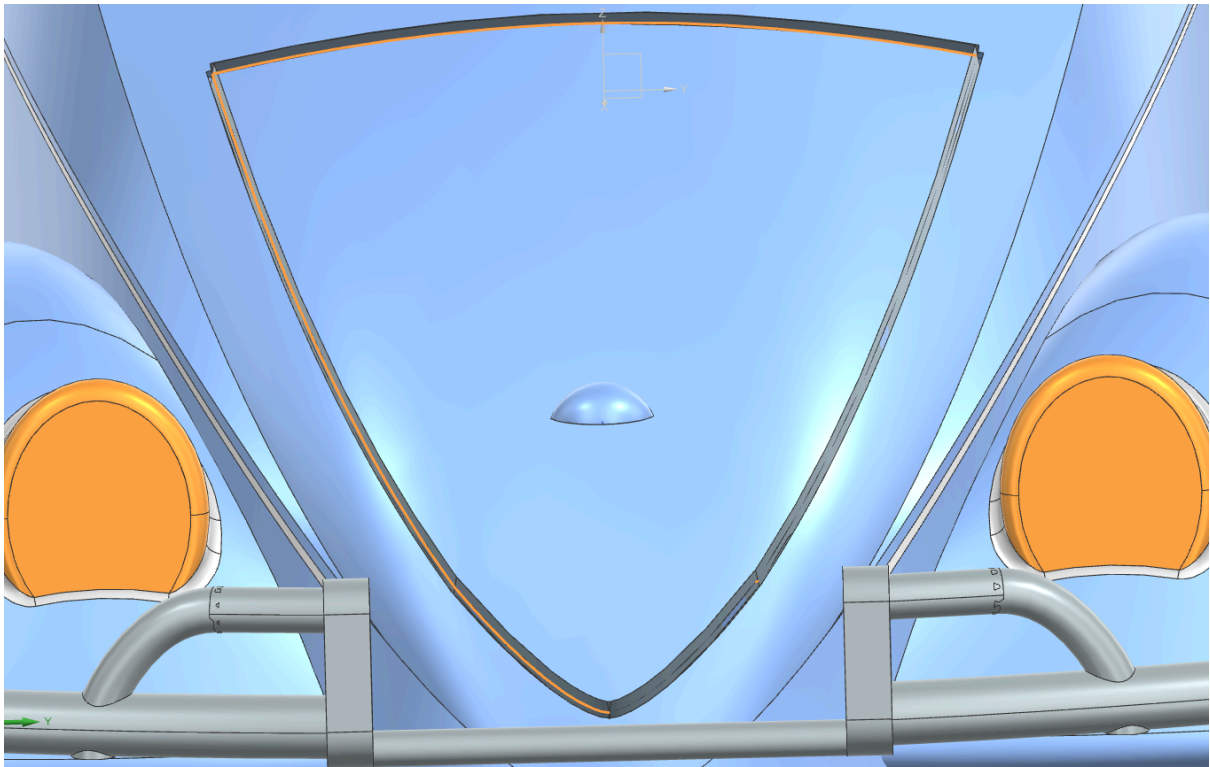
Finally, I drew a circle on a datum created at the center of the previous fixture and extruded it to the husk, combining it with the bumper and later combining everything to the husk.



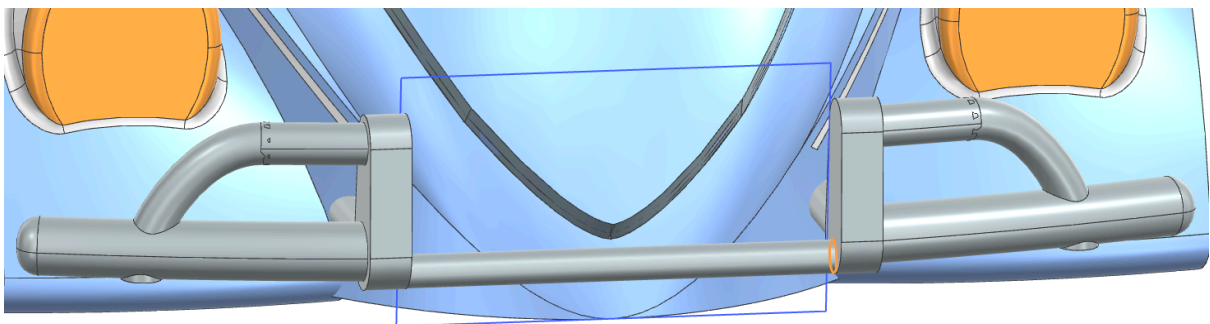
After that I moved on to the handle on the front hood. I made a spline on the Z-X plane touching the white fixture on the hood and then extended it slightly into the hood with tangent straight lines in order to be able to combine the handle to the husk. I then created a plane perpendicular to the spline at the connection point on the hood and drew a guiding circle. After that I created a surface from the circle and guiding line. Again, I thickened that and combined everything.



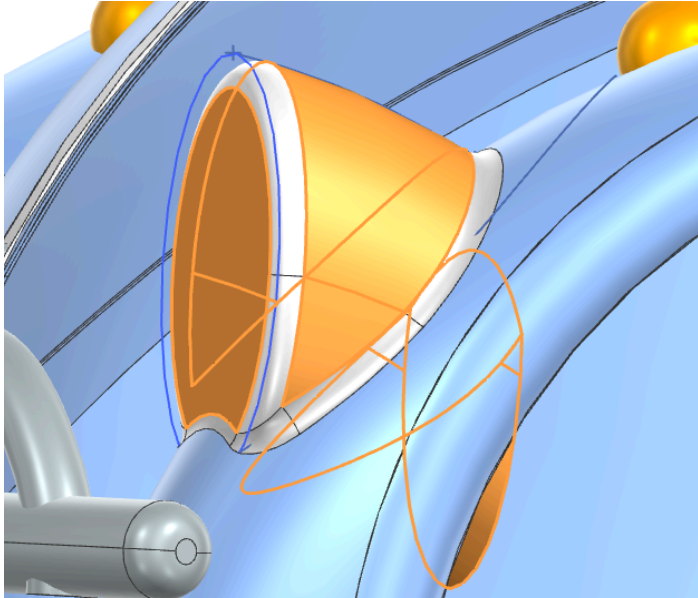
I then moved on to the back vents on the back hood. I drew a guiding spline on the ZY plane (not the same pitch as the windows, so I couldn't offset the curve on face) and projected it onto the back hood. I then drew the hole outline on the top plane and projected that onto the hood. I then created a surface from it, made a pattern along the line to both sides, thickened them all and subtracted them from the husk.



I then made the cut for the back trunk of the car. I again drew an outline by including the lines of the side part of the main husk and bottom line of the back window and offsetting them to make a “triangle”. I then projected that onto the back husk face, extruded it, thickened it by a small amount and subtracted that from the husk to make the outline of the trunk.



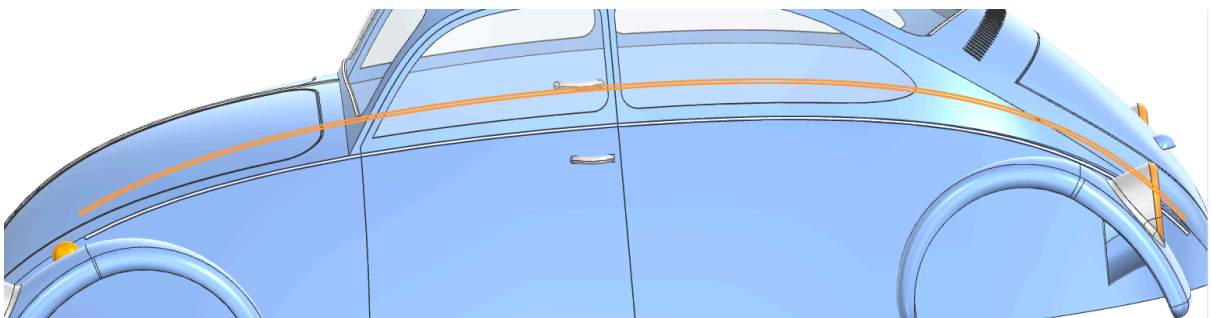
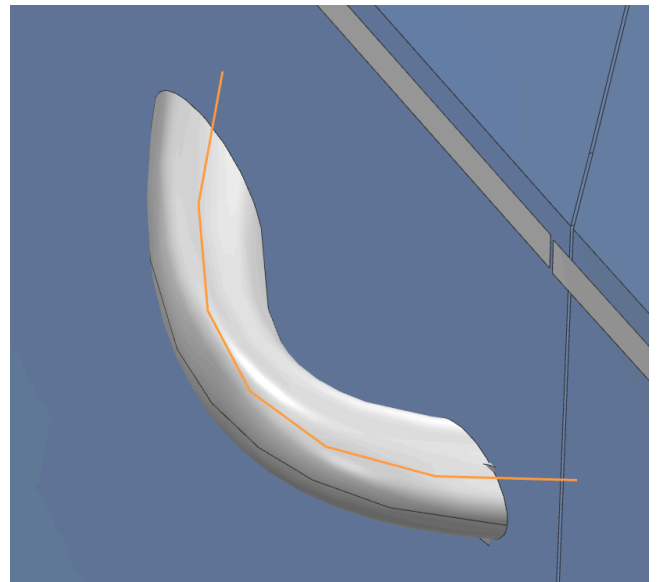
I then moved onto the back bumpers. I mirrored the front bumpers through a plane that was slightly more towards the front side of the car than the middle plane, as the back bumper goes out less than the front bumper in the beetle. I then created a square outline that touches the middle features of the bumpers, extruded that, combined the surfaces to make a solid and then subtracted that from the bumper, to get rid of the middle part of it. I then created a circle at the bottom of the middle fixtures and extruded from one side to the other. This was done as seen in some models of the beetle and also for functionality, as you wouldn't be able to open the trunk if the back trunk was the same as the front trunk.



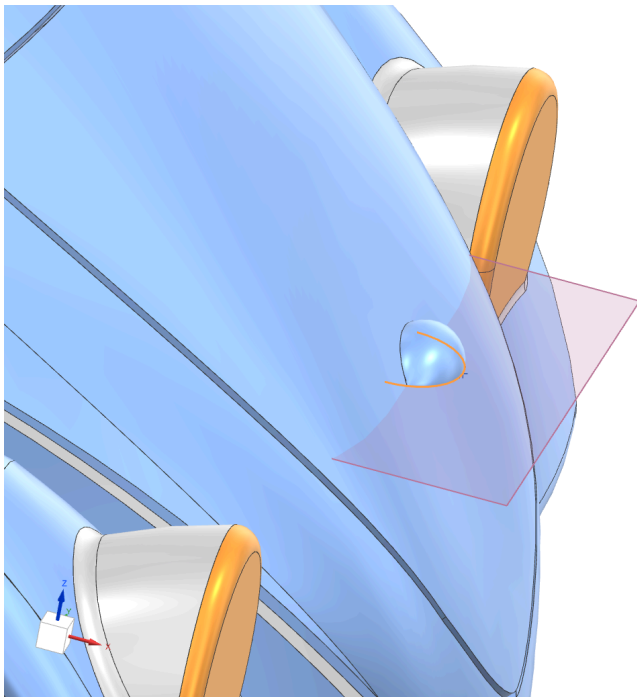
I then created a datum plane where I thought the headlights should start and drew an ellipse. I then made an arc connecting the center top of the ellipse and the wheel covers, going slightly beyond the surface of the cover in order to be able to combine it. I then made the surface from the ellipse and guiding curve, thickened it and combined everything.

With it united, I was able to create edge blends between the wheel cover and the headlight, making them look more realistic. I also added blends on the headlight itself. Finally, I mirrored it about the side plane and mirrored both about the same plane I mirrored the bumper, adding headlines to all 4 covers.

I then moved on to the door handles. I made through points where I thought sensible along the door and made a spline from them (again, making it go slightly further than the surface of the door in order to be able to unite everything. I then drew the circle on a datum perpendicular to the spline at the contact point, made the surface, and thickened it. I then mirrored that along a datum going through the middle of the door to make the inside handle and then mirrored both of these along the middle plane to add them to the door on the other side, uniting everything later.

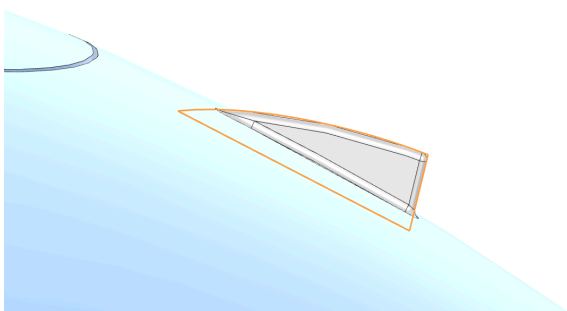
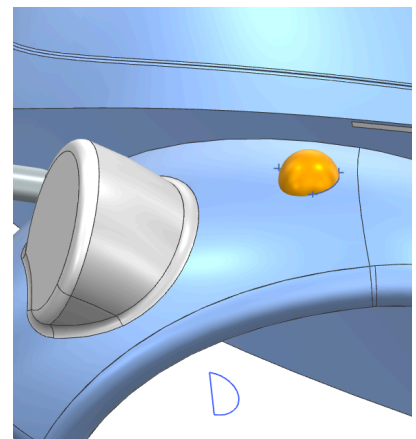


For the side stripes, I included the side edge of the husk and door on a drawing on the middle plane, slightly offset them and then closed the shape with lines. I then projected that onto the husk and the door, extruded it, filled surface on the end of the extrude, and thickened that filled surface. I then mirrored all and united all to the door and husk.



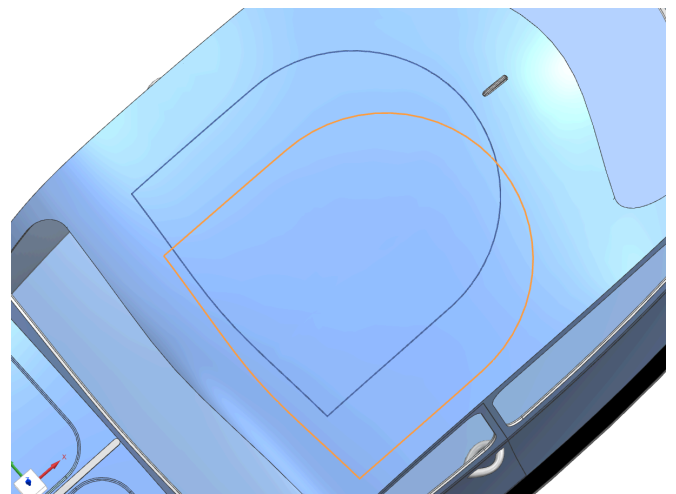
I then moved onto the handle of the trunk. I created a datum plane where I thought would be a good place for it, drew a semi circle, revolved it by 150 degrees (just enough to touch the trunk but not go through it on the other side), thickened it and united it to the husk.

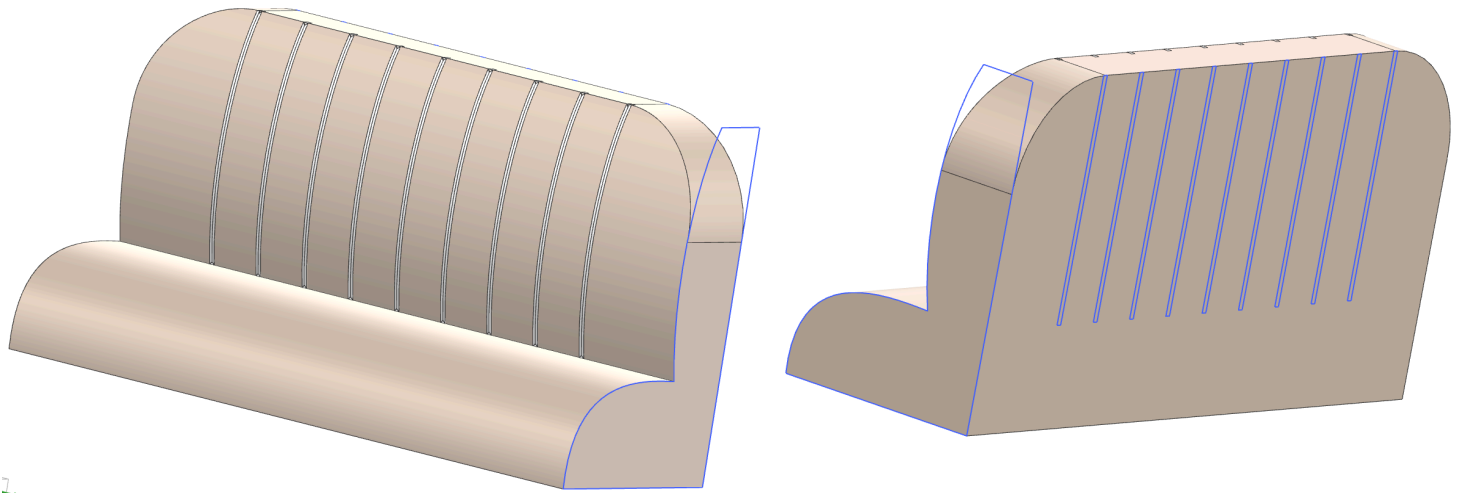
Then for the signaling lights, I did the same procedure as the trunk handle. I created a semi-circle on the bottom plane, projected that onto the wheel cover and revolved it as a solid, uniting everything. I then mirrored that to the other side.



I then made this sketch on the middle plane to make the vortex generator on the top of the car that some beetles have. I again went slightly through the husk in order to be able to unite everything. I extruded it as a solid and a unite. I then added edge/face blends to make it look more realistic.

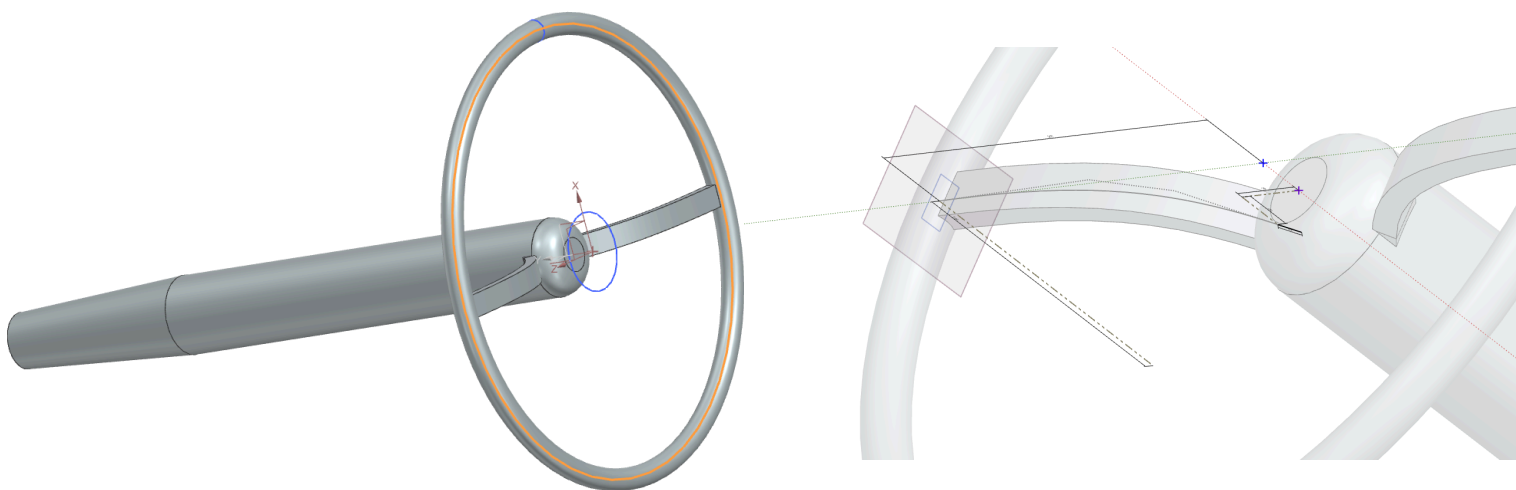
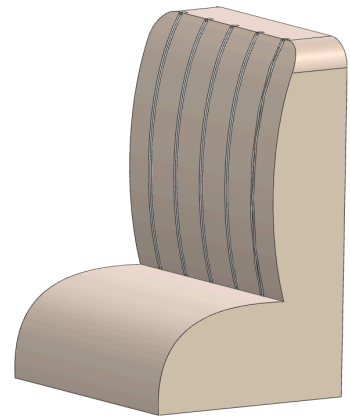
Finally, I drew the outline of the sunroof on the bottom plane by including the top outline of the windshield and sides of the husk, offsetting them and completing everything with a half circle. I projected that onto the top of the husk, extruded, thickened slightly and subtracted it from the husk.





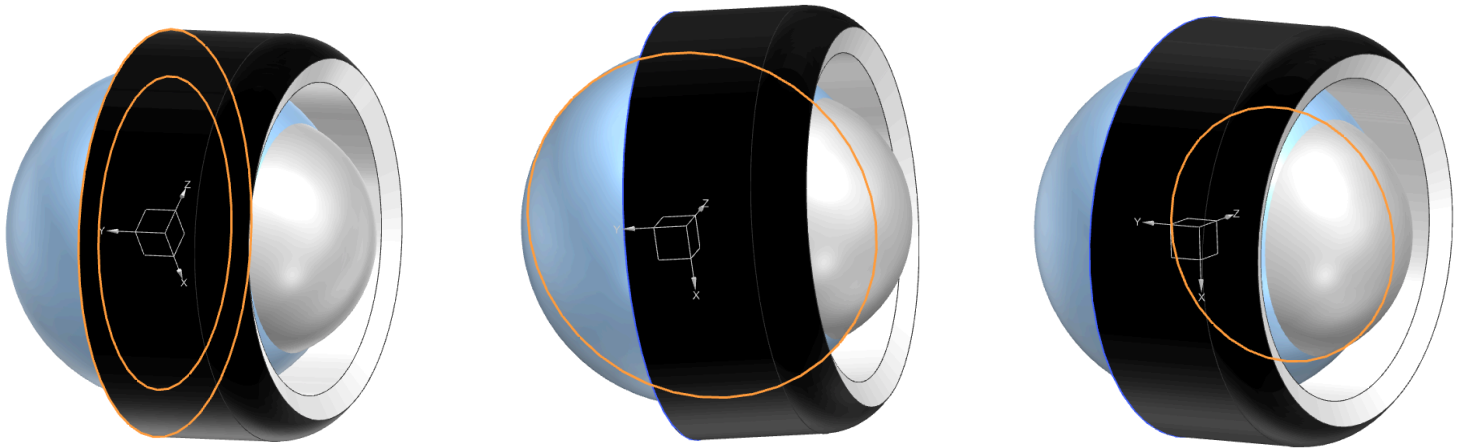
For the back seats, I drew out an outline with straight lines and splines arbitrarily, extruded that as a solid and added edge blends on the the side to make it look like an original seat of the beetle. After that, I drew a pattern on the back of the seat for the stripes. I projected that onto the front part, extruded that, thickened that, subtracted it from the seat and created again another of the same solid. I united that, but painted it white to make the stripes.

For the front seats, I copied the back seat exactly but made the extrude and the edge blend of the seat smaller. I then drew the same guides for the stripes, but with smaller intervals and used the same method to make them.

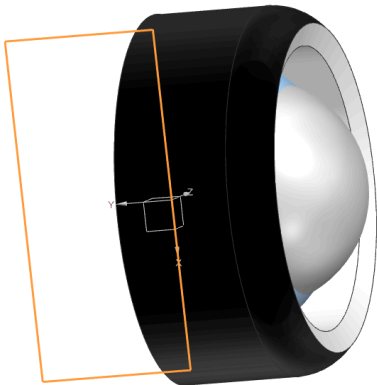


For the drive wheel, I started with a circle on the front plane and smaller circle through it to make the main part of the wheel, revolving that. I then drew another circle at the middle of the front plane and extruded it with an offset to make the shaft. I added edge blends and a

chamfer to it in order to make the wheel look as close as possible to the original wheel of the beetle. I then made a spline connecting the wheel to the shaft in the same shape as the beetle wheel connection and a square outline going through its middle. Again, the spline went into both solids in order to be able to unite everything. I then made a surface from it, thickened it and united all solids.

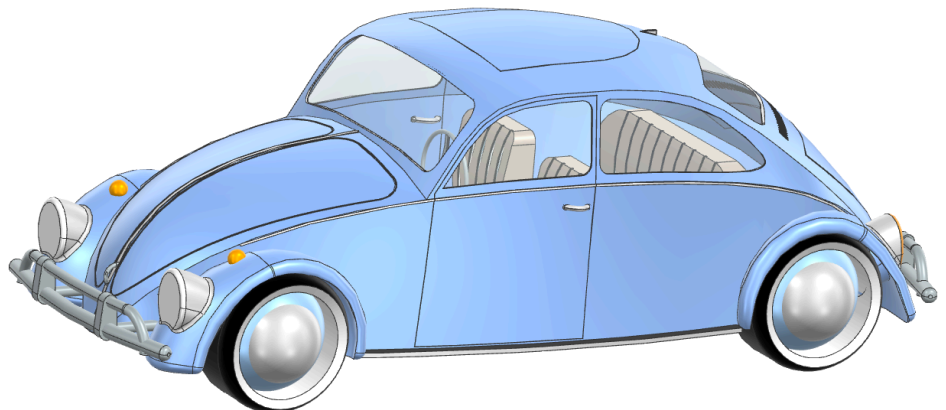


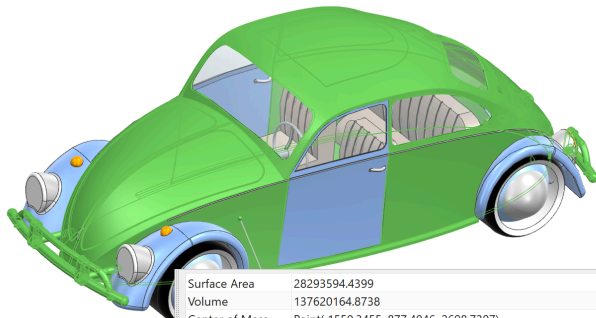
For the wheels, I started with two circles to make the rim. I extruded it and added an edge blend to the end as well as a chamfer on the inner edge, making it resemble the original wheel of the beetle. I then created the inner core sphere by creating a circle on the side plane with the same radius as the inner circle of the rim, revolving it and uniting it. Finally, I made a smaller offset circle to mimic the fixture of the original wheels, revolving and uniting it as well.



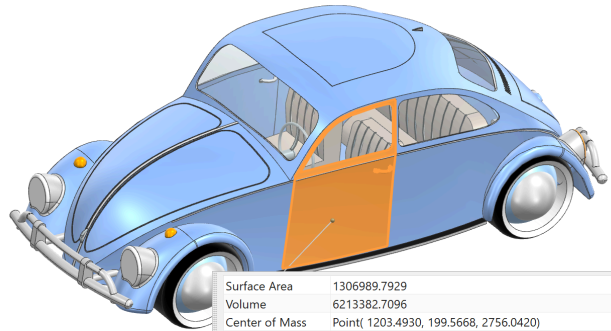
Finally, I drew a square touching the rim and going over the circles that were “poking out” from the back. I subtracted it from the body to remove the protuberation.

I then assembled everything together, having the husk as the main body and adding concentric mates for the wheels and touch/distance mates for the seats and drive wheel.

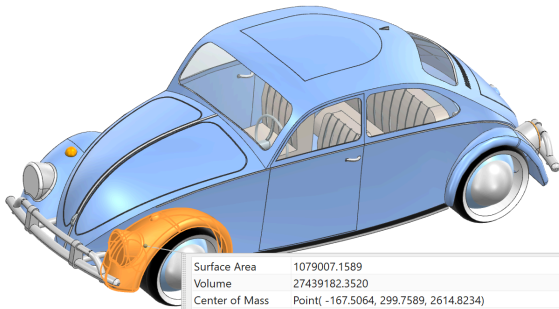




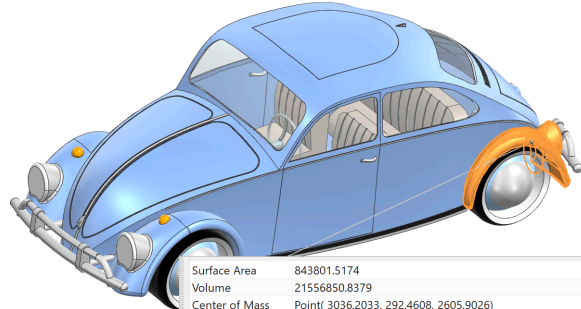
Surface Area	28293594.4399	mm ²	
Volume	137620164.8738	mm ³	
Center of Mass	Point(1550.3455, 877.4046, 2698.7307)	mm	
Mass	1077.6540	kg	
Weight	10568.1753	N	
Moments of Inertia	{ 9060930594.7661, 11997563037.2538, 5005638092.4951}	kg-mm ²	
Radii of Gyration	{ 2899.6577, 3336.6208, 2155.2123}	mm	
Principal Axes (Xp)	Vector(-0.0196, -0.0197, 0.9996)		
Principal Axes (Yp)	Vector(-0.0070, 0.9998, 0.0195)		
Principal Axes (Zp)	Vector(-0.9998, -0.0066, -0.0197)		
Principal Moments	{ 1586279754.1806, 1558673357.9078, 382078248.8669}	kg-mm ²	



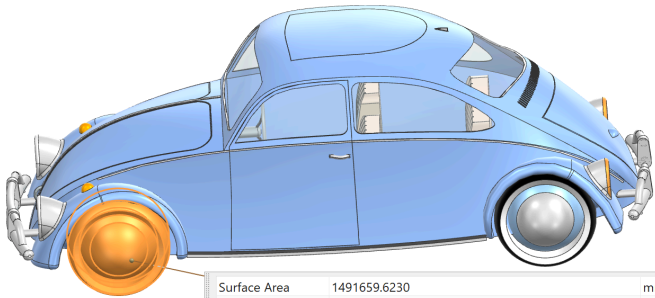
Surface Area	1306989.7929	mm ²	
Volume	6213382.7096	mm ³	
Center of Mass	Point(1203.4930, 199.5668, 2756.0420)	mm	
Mass	48.6548	kg	
Weight	477.1402	N	
Moments of Inertia	{ 374419739.3158, 445546370.1039, 75032950.8194}	kg-mm ²	
Radii of Gyration	{ 2774.0653, 3026.1035, 1241.8334}	mm	
Principal Axes (Xp)	Vector(0.0676, 0.9977, -0.0085)		
Principal Axes (Yp)	Vector(0.8663, -0.0629, -0.4955)		
Principal Axes (Zp)	Vector(-0.4949, 0.0262, -0.8686)		
Principal Moments	{ 5516655.0275, 3033302.0980, 2490338.6743}	kg-mm ²	



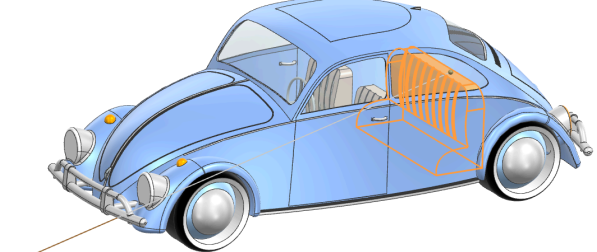
Surface Area	1079007.1589	mm ²	
Volume	27439182.3520	mm ³	
Center of Mass	Point(-167.5064, 299.7589, 2614.8234)	mm	
Mass	214.8664	kg	
Weight	2107.1192	N	
Moments of Inertia	{ 1494208808.0162, 1495782219.8130, 44094703.9188}	kg-mm ²	
Radii of Gyration	{ 2637.0686, 2638.4567, 453.0113}	mm	
Principal Axes (Xp)	Vector(0.1490, 0.9869, 0.0626)		
Principal Axes (Yp)	Vector(-0.1998, -0.0319, 0.9793)		
Principal Axes (Zp)	Vector(0.9684, -0.1585, 0.1924)		
Principal Moments	{ 21055831.9938, 19286189.3357, 4860274.7625}	kg-mm ²	



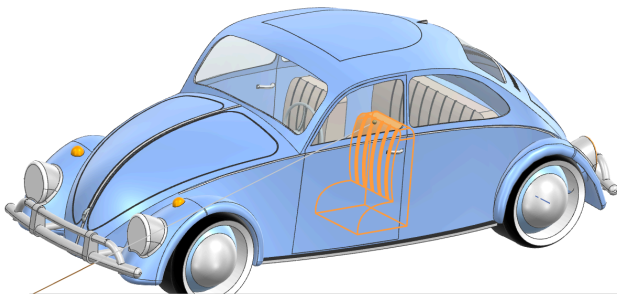
Surface Area	843801.5174	mm ²	
Volume	21556850.8379	mm ³	
Center of Mass	Point(3036.2033, 292.4608, 2605.9026)	mm	
Mass	168.8039	kg	
Weight	1655.4011	N	
Moments of Inertia	{ 1165097440.1540, 2713361707.7091, 1580010639.3643}	kg-mm ²	
Radii of Gyration	{ 2627.1800, 4009.2446, 3059.4173}	mm	
Principal Axes (Xp)	Vector(-0.1957, 0.9736, 0.1177)		
Principal Axes (Yp)	Vector(0.3583, -0.0408, 0.9327)		
Principal Axes (Zp)	Vector(0.9129, 0.2247, -0.3408)		
Principal Moments	{ 11352791.1867, 10261734.9336, 3126677.6346}	kg-mm ²	



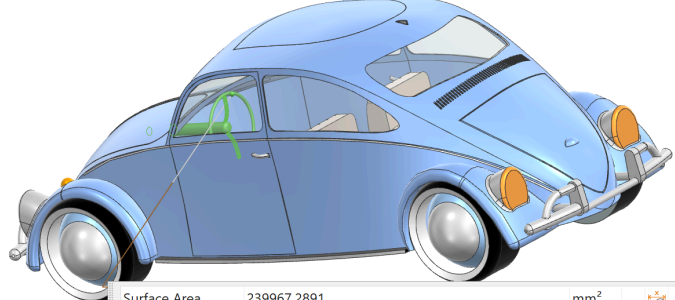
Surface Area	1491659.6230	mm ²	
Volume	84391690.6917	mm ³	
Center of Mass	Point(62.9946, 236.7198, 2451.9148)	mm	
Mass	660.8409	kg	
Weight	6480.6359	N	
Moments of Inertia	{ 4031037257.2026, 4009069311.8500, 60769559.1784}	kg-mm ²	
Radii of Gyration	{ 2469.7895, 2463.0505, 303.2456}	mm	
Principal Axes (Xp)	Vector(-0.0097, 0.9995, 0.0307)		
Principal Axes (Yp)	Vector(1.0000, 0.0097, 0.0)		
Principal Axes (Zp)	Vector(-0.0003, 0.0307, -0.9995)		
Principal Moments	{ 33559105.4348, 21104354.7884, 21104354.7884}	kg-mm ²	



Surface Area	2761832.7453	mm ²	
Volume	158013942.0762	mm ³	
Center of Mass	Point(2339.5995, 863.6052, 2638.7289)	mm	
Mass	1237.3503	kg	
Weight	12134.2613	N	
Moments of Inertia	{ 9725469789.5030, 15441782080.6000, 7855508501.3680}	kg-mm ²	
Radii of Gyration	{ 2803.5542, 3532.6644, 2519.6535}	mm	
Principal Axes (Xp)	Vector(0.9383, 0.0, -0.3458)		
Principal Axes (Yp)	Vector(0.3458, 0.0, 0.9383)		
Principal Axes (Zp)	Vector(0.0, -1.0000, 0.0)		
Principal Moments	{ 191398942.7904, 155462352.3803, 53331246.5200}	kg-mm ²	



Surface Area	1264578.4560	mm ²	
Volume	60657740.3601	mm ³	
Center of Mass	Point(1640.7993, 541.8956, 2619.7968)	mm	
Mass	474.9889	kg	
Weight	4658.0502	N	
Moments of Inertia	{ 3424297963.9752, 4560550443.9290, 1431210658.0408}	kg-mm ²	
Radii of Gyration	{ 2684.9984, 3098.6098, 1735.8414}	mm	
Principal Axes (Xp)	Vector(0.9440, 0.0, -0.3300)		
Principal Axes (Yp)	Vector(0.0, 1.0000, 0.0)		
Principal Axes (Zp)	Vector(0.3300, 0.0, 0.9440)		
Principal Moments	{ 26458787.1667, 21766361.4636, 11303972.0254}	kg-mm ²	



Surface Area	239967.2891	mm ²	
Volume	2555685.7506	mm ³	
Center of Mass	Point(785.2521, 538.5235, 3034.9988)	mm	
Mass	20.0127	kg	
Weight	196.2571	N	
Moments of Inertia	{ 190283370.2500, 197456027.4842, 18913736.3842}	kg-mm ²	
Radii of Gyration	{ 3083.5292, 3141.1078, 972.1568}	mm	
Principal Axes (Xp)	Vector(0.0, 1.0000, 0.0)		
Principal Axes (Yp)	Vector(0.0, 0.0, 1.0000)		
Principal Axes (Zp)	Vector(1.0000, 0.0, 0.0)		
Principal Moments	{ 774889.3322, 769696.3183, 138631.1289}	kg-mm ²	

With all of the volumes, and using 4 wheel covers, 4 wheels, 2 front seats, 1 husk, and 1 back seat, adding it all up yields 804957693 mm³ or 0.80495769 m³. This number is not agreeable with a normal car, but that is due to the governing dimensions and points given by the instructions, which were too small for a car.

